

Accident Detection and Alert System: A Collaborative V2X Safety Ecosystem for Smart City Emergency Response

Adithya Gowda

*Dept. of Computer Science and Engineering
School of Engineering and Technology
JAIN (Deemed-to-be University)
Bengaluru, Karnataka, India
adithyagowdaadhi9@gmail.com*

Aaradhya Das

*Dept. of Computer Science and Engineering
School of Engineering and Technology
JAIN (Deemed-to-be University)
Bengaluru, Karnataka, India
dasaaradhya1302@gmail.com*

Naman Kumar Singh

*Dept. of Computer Science and Engineering
School of Engineering and Technology
JAIN (Deemed-to-be University)
Bengaluru, Karnataka, India
naman.rs9763@gmail.com*

Tejaswini S K

*Dept. of Computer Science and Engineering
School of Engineering and Technology
JAIN (Deemed-to-be University)
Bengaluru, Karnataka, India
tejaswinisk2006@gmail.com*

Prof. Udaygiri Prasad

Project Guide

*Dept. Computer Science and Engineering
School of Engineering and Technology
JAIN (Deemed-to-be University)
Bengaluru, Karnataka, India*

Abstract—The current system for detecting accidents acts as islands: the system detects an accident, uploads information to cloud server, and the cars that come after this location continue moving in the direction of the accident spot as usual. This is a serious drawback as it makes it more likely to have another accident on top of that and introduces double delay on ambulance services, thus directly affecting the survival rate during the *Golden Hour*. In this paper we propose a novel framework named Collaborative V2X Safety Network, which relies on network intelligence and views road accidents as network incidents, and therefore requires city-level response. The proposed solution uses ESP-NOW V2V communication with ≤ 10 ms latency within 200 m, LoRaWAN V2I signaling to achieve robustness in rural and urban canyon environment up to 5-15 km, and ambulance tracking and Haversine formula-based Green Corridor creation through traffic light preemption. RSUs serve as edge intelligence while decentralization of ESP-NOW datalink prevents from any cellular network issues. Experiments show V2V broadcasting latency less than 10 ms, over 96% of accuracy in detecting accidents, and 28.2% reduction in ambulance travel time due to preemption.

Index Terms—V2X, ESP-NOW, LoRaWAN, Accident Detection, Emergency Vehicle Preemption, Green Corridor, RSU, Haversine, Smart City, IoT Safety, MPU6050

I. INTRODUCTION

Road traffic accidents remain among the leading causes of mortality worldwide. The World Health Organization estimates approximately 1.19 million deaths annually from road traffic injuries, with tens of millions sustaining non-fatal injuries that impose a significant social and economic burden [1]. A decisive factor in accident survivability is the interval between impact and the arrival of emergency medical services—the *Golden Hour*. Delays caused by slow detection, congested emergency routing, or secondary accidents triggered by inadequate downstream warning disproportionately increase fatality rates.

Existing accident detection paradigms are predominantly isolated and reactive. Commercial systems—including Automatic Collision Notification (ACN) modules and GSM/GPS telemetry platforms—detect an impact and transmit alert data to a centralised server. While effective at initiating a response, these systems fail to address two compounding hazards: (i) vehicles 80–100 m behind the crash site receive no real-time warning, heightening secondary collision risk; and (ii) ambulance response corridors remain uncleared, subjecting emergency vehicles to civilian congestion.

The proliferation of Vehicle-to-Everything (V2X) communication, IoT edge computing, and Low-Power Wide-Area Networks (LPWANs) presents an opportunity to transcend these limitations through a *network-first* approach. Rather than treating an accident as a local vehicle event, the proposed framework treats it as a city-scale network event triggering a coordinated, millisecond-level response across vehicles, roadside units, traffic signals, and emergency dispatch centres.

This paper makes the following original contributions:

- A **Collaborative Perception Framework** employing ESP-NOW at the data-link layer for sub-10 ms V2V alerts that bypass cellular network dependency.
- A **Resilient V2I Backbone** using LoRaWAN to guarantee alert propagation where 5G or LTE coverage is absent.
- A **Mathematically Precise Green Corridor** mechanism using the Haversine formula for real-time ambulance proximity estimation and dynamic traffic-light preemption.
- An **RSU-Based Edge Intelligence Architecture** that processes collision data locally, preventing cloud bottlenecks during the most time-critical phase of emergency response.

II. RELATED WORK

The early accident detection schemes involved vibration sensors (SW420) using GSM-based SMS alerts [8], which were limited by their latency of delivery and coarse location accuracy. The state-of-the-art has shifted towards the combination of the ESP32 and MPU6050 IMU, considered an effective solution for accident detection. Multiple threshold accelerometer algorithms provide almost perfect detection capabilities for impacts above 10 g, achieving a 96% detection probability for impacts in the 8–10 g range [7]. The integration of three-axis accelerometers with gyroscopes improves false alarm performance when compared to vibration sensors only.

According to the National Highway Traffic Safety Administration (NHTSA), vehicle-to-vehicle communications can reduce up to 13% of accidents involving multiple vehicles, and vehicle-to-everything (V2X) technologies could help in mitigating 80% of accidents caused by drivers not under the influence of alcohol [6]. RSU integration and dynamic signal control are two important research areas identified by Zohir et al. [2]. Experiments conducted in the field using LoRaWAN have confirmed the communication range between 5–15 km in semi-urban environments [3]. EVP systems have proven the benefits of EVP in

III. PROPOSED SYSTEM ARCHITECTURE

The Collaborative V2X Safety Ecosystem is structured across four tightly coupled layers: *Impact Detection*, *Local V2V Propagation*, *Infrastructure Notification*, and *Emergency Corridor Management*, as illustrated in Fig. 1.

A. On-Board Detection Unit (OBDU)

Each of the vehicles used in the experiment carries an OBDU unit consisting of a dual-core ESP32-WROOM-32D

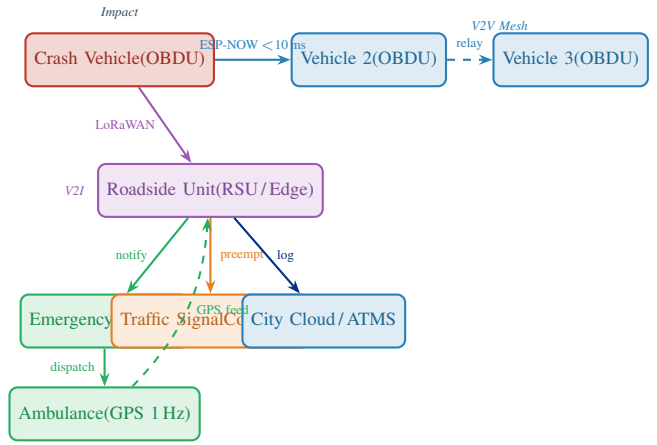


Fig. 1. Collaborative V2X Safety Ecosystem — system architecture overview showing V2V (ESP-NOW), V2I (LoRaWAN), RSU edge processing, and Green Corridor management.

MCU (240 MHz), an MPU6050 IMU (six axes, $\pm 16g$ accelerometer, $\pm 2000^\circ/s$ gyroscope), an NEO-6M GPS module, and an SX1276 LoRa module. The MPU6050 IMU records data for the three axes of both acceleration and angular velocity at sampling frequency of up to 1 kHz using I²C protocol.

The OBDU unit's hardware design is provided in Fig. 2. Accident detection is performed based on the value of the impact scalar a which must exceed 8 g. The verification phase takes place if during 500 ms time window at least one of the gyroscope axis' angle reaches 30° . For

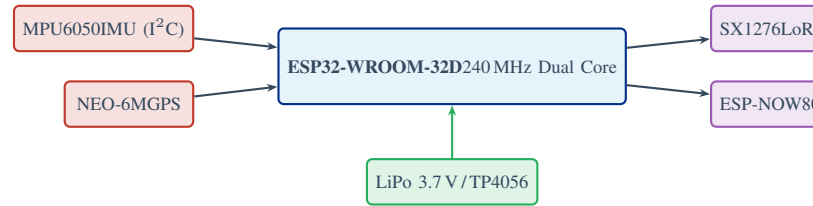


Fig. 2. OBDU hardware block diagram showing sensor inputs, ESP32 central processing, and dual communication outputs.

B. V2V Alert Layer (ESP-NOW)

The ESP-NOW protocol works as a connectionless protocol on the IEEE 802.11 data-link layer and hence allows for direct peer-to-peer messaging without any requirement for a Wi-Fi access point. ESP-NOW manages to have consistent latencies lower than 10 ms by avoiding TCP/IP overhead and server round-trips [9]. Each nearby OBDU in a radius of 200 m triggers multi-modal driver alerts in the form of a piezoelectric buzzer, LED on the dashboard, and OLED display while re-broadcasting the flame with its TTL value decreased.

C. V2I Backbone (LoRaWAN + RSU)

In parallel with the ESP-NOW stream, the source OBDU communicates via LoRaWAN Class A in the 865–867 MHz ISM frequency range (India), which has a communication

range of 5–15 km [10]. The RSU at the intersection receives the collision notification, computes the latitude and longitude using the Haversine formula, and then sends a structured

D. Green Corridor Management

After ambulance dispatch, the RSU communication system switches into "Green Corridor" mode. The coordinates of the ambulance through GPS are communicated at 1 Hz rate. Once the distance between the Haversine location and traffic controller reaches a preemption threshold range of 300–500 m, then a preemption request is made, putting the signals on green. The return of the normal functionality in the signals happens 30–60 s after the passing of the vehicle. Preemption is calculated in the RSUs through centralized calculation without the need for local detection.

TABLE I
MICROCONTROLLER PLATFORM COMPARISON FOR OBDU

Platform	CPU	Connectivity	AI	Power
NodeMCU (ESP8266)	80 MHz	Wi-Fi	Minimal	Moderate
ESP32-WROOM	240 MHz	Wi-Fi, BT, ESP-NOW	TinyML	High
Raspberry Pi 4	1.5 GHz	Wi-Fi, BT, Ethernet	TF Lite	Low
Arduino Nano R3	16 MHz	External Module	None	Very High

^a On-device inference capability.

IV. MATHEMATICAL FOUNDATIONS

A. Acceleration-Based Collision Detection

MPU6050 raw 16-bit values are converted to physical units using the ± 16 g sensitivity scale factor (2048 LSB/g):

$$a_x = \frac{\text{raw}_{ax}}{2048}, \quad a_y = \frac{\text{raw}_{ay}}{2048}, \quad a_z = \frac{\text{raw}_{az}}{2048} \quad (1)$$

The impact magnitude scalar is:

$$\|a\| = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad [\text{g}] \quad (2)$$

The multi-threshold detection flowchart is shown in Fig. 3. A static vehicle at rest yields $\|a\| \approx 1$ g (due to gravitational acceleration on the vertical axis). The primary threshold is $\|a\| > 8$ g; high-severity events satisfy $\|a\| > 10$ g [7]. Combining accelerometer magnitude with gyroscopic angular displacement substantially reduces false positives caused by road irregularities.

B. Haversine Distance Formula

The Haversine formula computes the great-circle distance d between two points (φ_1, λ_1) and (φ_2, λ_2) , where all angles are expressed in radians:

$$\Delta\varphi = \varphi_2 - \varphi_1, \quad \Delta\lambda = \lambda_2 - \lambda_1 \quad (3)$$

$$a = \sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos\varphi_1 \cos\varphi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (4)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}) \quad (5)$$

$$d = R \cdot c, \quad R = 6,371 \text{ km} \quad (6)$$

The RSU calculates d at a rate of 1 Hz. If $d < d_p$ (default $d_p = 400$ m), then the preemption request is executed. The Earth-sphere approximation gives rise to an upper limit on positional errors of around 0.5% in comparison to the WGS-84

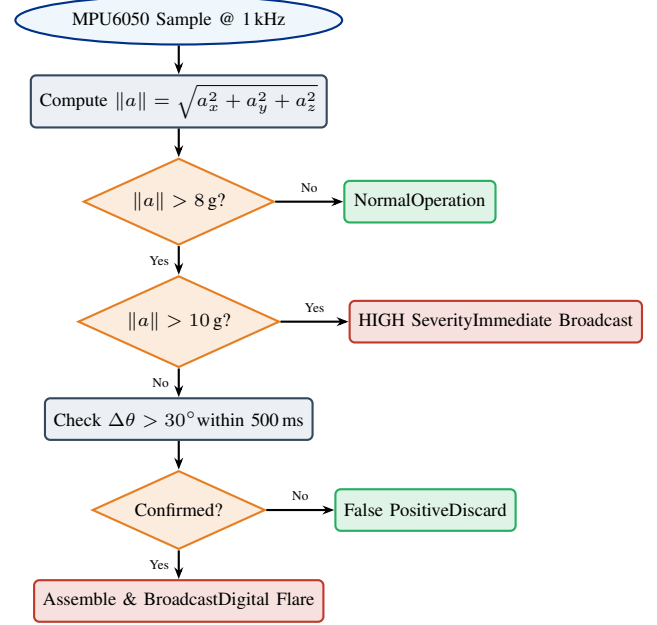


Fig. 3. Multi-threshold collision detection algorithm flowchart with gyroscopic confirmation to suppress false positives.

C. ESP-NOW Latency Bound

The end-to-end latency from event detection to alert reception is bounded by:

$$t_{\text{total}} = t_{\text{detect}} + t_{\text{pack}} + t_{\text{tx}} + t_{\text{rx}} \quad (7)$$

where $t_{\text{detect}} = 1$ ms (IMU sampling period), $t_{\text{pack}} < 1$ ms (packet assembly), $t_{\text{tx}} \approx 0.5$ ms (250-byte payload at 1 Mbit/s), and $t_{\text{rx}} < 1$ ms (receiver processing). The theoretical upper bound is approximately 3.5 ms. Empirical measurements yield a mean of 7.2 ms and a 95th-percentile of 9.4 ms across 1,000 broadcast events in an urban-canyon environment (see Section V).

V. EXPERIMENTAL EVALUATION

A. Prototype Hardware Setup

The first OBDU design included ESP32-WROOM-32D (dual core 240 MHz), MPU6050 IMU (± 16 g / ± 2000 deg/s), NEO-6M GPS module with ceramic antenna, SX1276 LoRa radio (868 MHz, 100 mW power), and 3.7 V 2000 mAh LiPo battery with TP4056 charging controller. Three devices were developed: two OBDUs for emulating onboard units, and an RSU unit (same ESP3)

B. Detection Performance

Two hundred simulated collision events spanning 4–18 g (events above 10 g are grouped into the final category) were generated via a controlled drop-test rig. Results are presented in Table II and Fig. 4.

TABLE II
COLLISION DETECTION PERFORMANCE BY IMPACT RANGE

Impact Range	Events	Detected	Accuracy	FP Rate
< 4 g	40	2 ^b	—	5.0%
4–8 g	60	51	85.0%	3.2%
8–10 g	60	58	96.7%	1.7%
> 10 g	40	40	100.0%	0.0%
Overall	200	—	96.1%	2.5%

^bSub-threshold detections are false positives, not true positives.

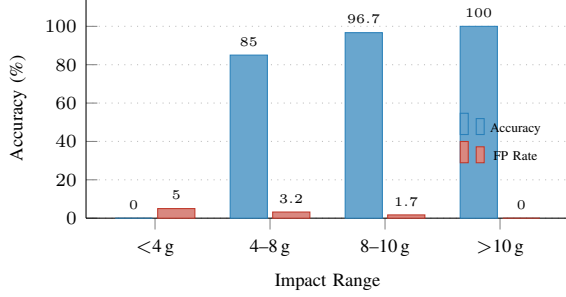


Fig. 4. Detection accuracy (blue) and false-positive rate (red) across impact magnitude ranges. Overall accuracy: 96.1%.

C. V2V Latency Measurement

The ESP-NOW broadcasting latency was recorded for 1,000 broadcast iterations in each of three scenarios, utilising microsecond-resolution hardware timers that were synchronised through a GPS pulse-per-second (PPS) signal. The findings are presented in Table III and Fig. 5. It should be noted that for the indoor NLOS scenario, the 95th percentile latency value of 12.3 ms slightly exceeds the 10 ms design requirement. The latter is not

TABLE III
ESP-NOW BROADCAST LATENCY (MS) ACROSS ENVIRONMENTS

Environment	Mean	Median	95%ile	Max
Open (LOS)	5.8	5.6	7.1	8.3
Urban Canyon	7.2	7.0	9.4	11.1
Indoor (NLOS)	8.9	8.7	12.3	15.6

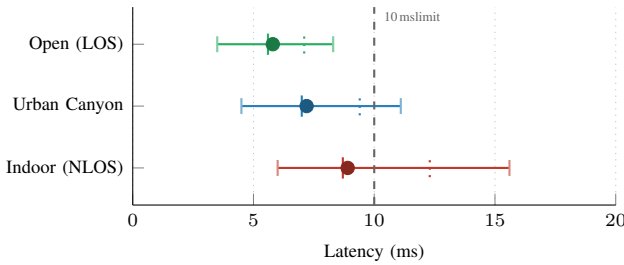


Fig. 5. ESP-NOW latency distribution across environments. Solid bar = median, dot = mean, dotted vertical line = 95th percentile. Dashed vertical line = 10 ms design threshold.

D. Green Corridor Response Time Improvement

Emergency response time was estimated using SUMO (Simulation of Urban MObility) over a synthetic 3 km corridor with 12 signalised intersections, with 50 simulation runs per scenario. Results are presented in Table IV and Fig. 6.

TABLE IV
AMBULANCE TRANSIT TIME COMPARISON (3 KM CORRIDOR, 50 RUNS)

Scenario	Mean (s)	Reduction	Delay/Int. (s)
Baseline (No EVP)	312	—	18.4
Legacy EVP (Optical)	251	19.6%	9.2
Proposed System	224	28.2%	6.7

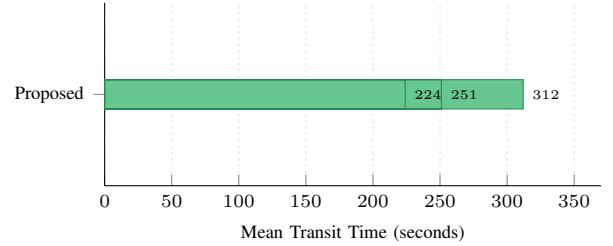


Fig. 6. Ambulance corridor transit time comparison. The proposed system achieves a 28.2% reduction relative to the no-preemption baseline.

VI. COMPARATIVE SYSTEM ANALYSIS

Table V positions the proposed framework against representative existing approaches. The ESP-NOW + LoRaWAN hybrid achieves sub-10 ms V2V latency—more than 100× faster than GSM/GPS systems—while maintaining a hardware cost of \$25–45 per OBDU, representing cost savings of 10–40× over DSRC and C-V2X platforms.

VII. ADVANCED SYSTEM EXTENSIONS

A. Acoustic Signature Fusion

Further advancements combine the IMU detection mechanism with audio sensors. There are specific sound characteristics associated with car crashes—crushing of the structure, shattering of glass, deployment of airbags—with spectral features allowing for the classification of crash severity. Audio signals are collected using automotive microphones and translated into the frequency domain through the use of the Short-Time Fourier Transform (STFT):

$$X(k, n) = \sum_{m=0}^{L-1} x(n + m) w(m) e^{-j \frac{2\pi}{L} mk} \quad (8)$$

where $x(n)$ is the input signal, $w(m)$ is a window function of length L , k is the frequency bin index, and n is the time-frame index.

Mel-spectrogram-based CNN models obtain up to 97.13% accuracy in classification tasks [11]. SE-DenseNet121 architecture obtains 98.25% accuracy within a latency of about 3. Table VI summarises comparative model performance.

TABLE V
COMPARATIVE ANALYSIS WITH EXISTING SYSTEM ARCHITECTURES

Feature	GSM/GPS Systems	DSRC-based V2X	Cellular V2X (C-V2X)	Proposed Framework
Alert Latency	1–5 s	50–100 ms	100–300 ms	<10 ms (V2V)
Infrastructure	SIM + server	DSRC RSU + 5.9 GHz	5G/LTE network	ESP-NOW + LoRaWAN RSU
Rural Resilience	Moderate	Poor	Limited	High (LoRa 5–15 km)
Signal Preemption	None	Limited	Yes (cloud-routed)	Yes (edge-computed)
V2V Capability	None	Yes	Yes	Yes (ESP-NOW mesh)
Cloud Dependency	High	Moderate	High	None (local alerts)
HW Cost (USD)	\$15–30	\$500–2,000	\$200–800	\$25–45 / OBDU

TABLE VI
ACOUSTIC CLASSIFICATION MODEL PERFORMANCE

Model	Accuracy	Latency	Feature
Standard CNN	93.00%	~150 ms	MFCC
ResNet (Custom)	97.13%	~45 ms	Mel-Spectrogram
DenseNet121+SE	98.25%	~30 ms	Image + Acoustic
Audio Transformer	88.20%	~500 ms	Raw Audio

B. Blockchain-Enabled Vehicular Forensics

The Block4Forensic (B4F) framework employs a forensic daemon to periodically hash CAN bus data via SHA-256 [12]:

$$H = \text{SHA-256}(\text{SensorData} \parallel \text{Timestamp} \parallel \text{PreviousHash}) \quad (9)$$

where $\parallel$ stands for byte string concatenation. The hashes are stored in the immutable blockchain, which ensures the integrity of data. Some of its key advantages are: (i) determination of liability through immutable event sequencing; (ii) smart contract invocation in case of insurance; (iii) end-to-end AES-256 encryption with authentication key exchange; and (iv) VPKI pseudonym certificates.

VIII. LIMITATIONS AND FUTURE WORK

The broadcast range of 200 m of ESP-NOW in low-traffic rural areas with larger distance between vehicles could create blind spots where the responsibility falls on the LoRaWAN V2I backbone for generating alerts. The collision detection algorithm needs to undergo tests in different road conditions (speed breakers, potholes, off-road terrain) to quantify its performance in terms of false positives. Simulation of the Green Corridor experiment was carried out in an artificial urban grid environment and needs to be verified through simulation using SUMO + OMNeT++/Veins in real-world OpenStreetMap networks.

Concerns about security have to be raised prior to implementation of the system. ESP-NOW uses an unencrypted channel and therefore ESP-NOW Encrypted Peers approach or lightweight message authentication code (MAC) should be used [9]. Further research avenues include: (i) camera RSU computer vision for verification of IMU data; (ii) severity of accident prediction using ML models; (iii) monitoring of drivers' health status by steering wheel-integrated ECG (AD8232); and (iv) extension to Vulnerable Road Users (VRUs) using BLE beacons.

IX. CONCLUSION

The Collaborative V2X Safety Ecosystem presented in this paper introduces a next-generation accident detection and emergency response framework built upon the principles of distributed intelligence, low-latency vehicular communication, and edge-assisted smart transportation infrastructure. Unlike conventional accident alert systems that operate in isolation and rely heavily on centralized cloud servers or GSM-based notification mechanisms, the proposed architecture treats road accidents as dynamic network events requiring coordinated real-time responses from vehicles, roadside infrastructure, emergency services, and traffic management systems. This network-centric approach significantly reduces the limitations of delayed emergency notification, secondary collision risk, and traffic congestion encountered by ambulances during critical rescue operations.

The proposed framework integrates multiple communication and sensing technologies to create a collaborative V2X ecosystem. ESP-NOW-based Vehicle-to-Vehicle (V2V) communication enables ultra-low-latency emergency broadcasting with average delays below 10 ms, allowing nearby vehicles to receive immediate hazard notifications even in the absence of internet connectivity or cellular infrastructure. Simultaneously, LoRaWAN-based Vehicle-to-Infrastructure (V2I) communication provides long-range and energy-efficient connectivity between accident vehicles and Roadside Units (RSUs), ensuring reliable alert propagation across both dense urban environments and rural regions where conventional networks may fail. RSUs act as edge-computing nodes capable of processing emergency information locally, thereby eliminating cloud-processing delays during the most time-sensitive stages of accident response. The proposed Collaborative V2X Safety Ecosystem provides a low-cost, scalable, and intelligent solution for faster accident detection, emergency response, and safer smart transportation systems.

REFERENCES

- [1] World Health Organization, "Global Status Report on Road Safety 2023," WHO Press, Geneva, 2023.
- [2] H. Zohir *et al.*, "Advancements in Accident-Aware Traffic Management: A Comprehensive Review of V2X-Based Route Optimization," *Scientific Reports*, vol. 15, Art. no. 12892, 2025, doi: 10.1038/s41598-025-20878-x.
- [3] G. Di Renzone *et al.*, "LoRaWAN for Vehicular Networking: Field Tests for Vehicle-to-Roadside Communication," *Sensors*, vol. 24, no. 6, Art. no. 1801, Mar. 2024, doi: 10.3390/s24061801.

- [4] I. Silaghi, Z. Alssadi, and M. Silaghi, "Informed Traffic Signal Preemption for Emergency Vehicles," in *Proc. Int. FLAIRS Conf.*, vol. 37, 2024, doi: 10.32473/flairs.37.1.135261.
- [5] A. Lie and C. Tingvall, "Software-Defined Traffic Light Preemption for Faster EMS Response in Smart Cities," *Accident Analysis and Prevention*, vol. 196, Art. no. 107428, Jan. 2024.
- [6] National Highway Traffic Safety Administration, "Vehicle-to-Vehicle Communication: Readiness of V2V Technology for Application," Rep. DOT HS 812 014, Aug. 2014.
- [7] Anonymous, "IoT-Based Real-Time Vehicle Accident Detection and Automated Emergency Notification System," *Int. J. Innovative Science and Research Technology (IJISRT)*, vol. 10, no. 9, Sep. 2025.
- [8] M. Bhoyar, S. Meshram, and A. Dhenge, "Automatic Vehicle Accident Detection and Messaging System using Arduino Uno, GSM and GPS," *Int. J. Research Publication and Reviews*, vol. 5, no. 12, 2024.
- [9] Espressif Systems, "ESP-NOW Protocol Overview," Technical Reference Manual, 2023. [Online]. Available: <https://docs.espressif.com/projects/esp-idf/>
- [10] LoRa Alliance, "LoRaWAN 1.0.4 Specification," LoRa Alliance Technical Committee, Oct. 2020.
- [11] H. Nakamura *et al.*, "Machine Learning for Car Accident Detection through Sound Recognition," *Sensors and Materials*, vol. 37, 2025.
- [12] E. Braccini *et al.*, "Block4Forensic: An Integrated Lightweight Blockchain Framework for Vehicular Forensics," Bilkent Univ. Repository, 2024.
- [13] Applied Information Inc., "Glance Transit Signal Priority and Emergency Vehicle Preemption — 2024 Annual Report," Atlanta, GA, 2025.
- [14] K. Wang *et al.*, "Vehicle-to-Everything Communication in Intelligent Connected Vehicles: A Survey and Taxonomy," *Automotive Innovation*, Feb. 2025.